

7 (a) $KE_{\max} = eV_s$
 $= (1.60 \times 10^{-19})(1.1)$
 $= 1.76 \times 10^{-19} \text{ J}$

(b)

$$f = \frac{c}{\lambda} = \frac{(3.00 \times 10^8)}{450 \times 10^{-9}}$$

$$hf = eV_s + \phi$$

$$\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{450 \times 10^{-9}} = (1.6 \times 10^{-19})(1.1) + \phi$$

$$\phi = 2.66 \times 10^{-19} \text{ J}$$

$$\phi = hf_0$$

$$f_0 = \frac{2.66 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$f_0 = 4.01 \times 10^{14} \text{ Hz}$$

- (c) The energy of the photons remains the same because as $E = hf$ and f is the same. Hence, by the photoelectric equation, the maximum kinetic energy of photoelectrons emitted will not be affected.

(d)

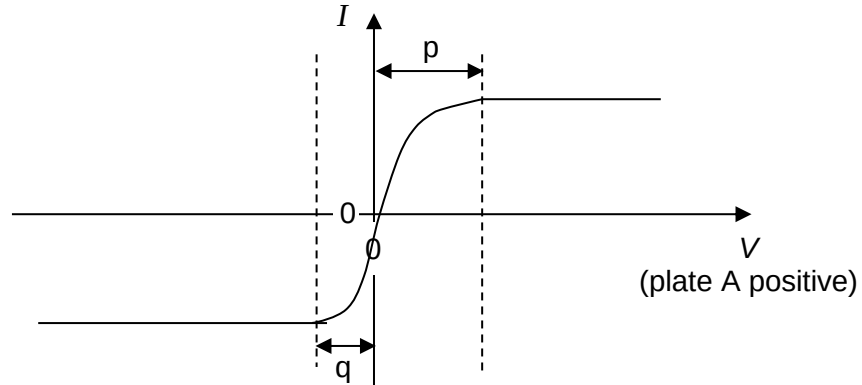


Fig. 7.3

Additional information:

Photoelectrons will be emitted from both plates.

Since plate A has a lower threshold frequency than plate B, photoelectrons emitted from plate A have higher maximum kinetic energy than plate B. Stopping potential of plate A, V_{SA} will be larger than that of plate B, V_{SB} .

